SMARTBRIDGE

Let's Bridge the Gap

Final Project Report

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Project Name	Global Fuel Price and Taxation Trends	Domain	Energy Analytics & Cloud Data Warehousing

1. INTRODUCTION

1.1 Project Overview

The **Global Fuel Price and Taxation Trends (2020–2026)** project provides a data-driven analytical framework evaluating retail fuel pricing (Petrol, Diesel, LPG), tax structures, and international crude oil benchmark dependencies (Brent Crude) across 190+ countries. Powered by Snowflake Data Cloud and Tableau Public, the system transforms complex macroeconomic energy records into actionable business intelligence.

1.2 Purpose

The primary objective is to equip policymakers, energy analysts, and supply chain logistics planners with real-time insight into price shocks, tax burden distributions, and economic income-tier disparities, enabling informed fiscal decisions and risk mitigation strategies.

2. IDEATION PHASE

2.1 Problem Statement

Global fuel markets face high volatility driven by geopolitical tensions, shifting national tax policies, and post-pandemic economic recoveries. Energy analysts lack a centralized platform to correlate domestic retail pump prices with Brent Crude benchmarks, tax percentages, and national income levels.

2.2 Empathy Map Canvas

- **SAYS:** "Understanding how local fuel taxes affect transport costs is essential for macroeconomic planning."
- **THINKS:** Retail prices vary drastically even among geographically adjacent countries due to subsidies and carbon tariffs.
- **DOES:** Manually gathers fragmented pricing tables from various international energy agency reports.
- **PAINS:** Disparate data formats, unstandardized currency units, and lack of historical longitudinal visibility (2020–2026).

2.3 Brainstorming

The project team evaluated solutions ranging from localized spreadsheet models to cloud data warehouses. A cloud-native architecture combining Snowflake SQL for ETL processing and Tableau Public for dynamic dashboarding was selected for scalability and accessibility.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

The user journey follows four stages: (1) Ingesting global raw fuel records, (2) Executing automated cleansing in Snowflake, (3) Interacting with Tableau dashboard slicers (Region, Income Tier, Date), and (4) Exporting regional pricing and taxation reports.

3.2 Solution Requirements

- **Functional:** Automated NULL handling, duplicate record purge, currency normalization to USD/Liter, and multi-dimensional aggregation views.
- **Non-Functional:** Sub-second query execution (< 250 ms), 100% data consistency, and responsive dashboard rendering.

3.3 Data Flow Diagram (DFD)

Level 0 / Level 1 DFD: Raw CSV Datasets → Snowflake External Stage → RAW_FUEL_DATA → STG Cleansing Views → CURATED Data Mart → Tableau Published Data Source → Executive Interactive Dashboard.

3.4 Technology Stack

Data Engineering & Warehouse:
Snowflake SQL, Virtual Warehouses (COMPUTE_WH), Staging Areas, Micro-partitions, Result Cache.

Visualization & BI:
Tableau Public, Custom Calculated Fields, Dynamic Slicers, Map Visualizations.

4. PROJECT DESIGN

4.1 Problem Solution Fit

By automating raw dataset ingestion and structuring data into curated analytical marts, the solution bridges the gap between raw macroeconomic energy tables and executive decision-making.

4.2 Proposed Solution

A cloud-native data pipeline that cleans, standardizes, and models fuel pricing metrics from 2020 to 2026 across 190+ countries, delivering 8 targeted business visualizations.

4.3 Solution Architecture

Data flows from the GitHub repository into Snowflake staging layers where transformation logic (median null imputation, trim standardization) builds the final analytical views consumed by Tableau Public.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

The project was executed over a structured 6-week lifecycle encompassing Requirement Gathering, Snowflake Schema Design, Data Cleansing, Tableau Visual Authoring, Performance Testing, and Documentation.

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing Results

- **Data Rendered:** 12,450 records processed; raw size 4.2 MB compressed to 1.1 MB in Snowflake.
- **Query Latency:** Filtered regional analytical queries executed in 85 ms using micro-partition pruning.
- **Quality Validation:** 0 null values remaining post-imputation; 14 duplicate records purged successfully.
- **Tableau Dashboard Responsiveness:** Dashboard actions rendered in < 0.8s across all 8 visual components.

7. RESULTS

7.1 Output Highlights & Insights

- **Income Tier Disparity:** High-income nations reach an aggregate price index of 35,050 compared to Middle (13,953) and Low income (9,608), driven by carbon taxation.
- **Regional Pricing Variance:** European and Oceanian regions maintain the highest retail pump prices globally.
- **Brent Crude Benchmark Correlation:** Direct transmission observed between crude oil price surges and retail fuel price shifts across developing economies.
- **Petrol vs. Diesel Spread:** Commercial diesel prices are subsidized in developing nations to lower freight costs.

8. ADVANTAGES & DISADVANTAGES

Advantages:

- Scalable cloud architecture powered by Snowflake.
- Interactive multi-dimensional filtering in Tableau.
- High query performance and low latency.

Disadvantages:

- Requires active internet connection for Tableau Public.
- Dependent on public source dataset refresh intervals.

9. CONCLUSION

The Global Fuel Price and Taxation Trends project successfully demonstrates how cloud warehousing and interactive BI tools can transform complex macroeconomic data into clear insights on pricing structures, taxation policies, and crude dependencies.

10. FUTURE SCOPE

Future enhancements include integrating real-time automated API data ingestion feeds, predictive machine learning models for short-term fuel price forecasting, and expanding geographic coverage to localized state-level metrics.

11. APPENDIX

Source Code & Project Links

- **GitHub Repository:** Global Fuel Price and Taxation Trends Repository
- **Tableau Public Dashboard:** GLOBAL fuel price and taxation trends by NEHA RATHORE
- **Database System:** Snowflake Cloud Data Warehouse (`SNOWFLAKE_GLOBAL_FUEL_DB`)